Final Homework — Introduction to Data Science

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Final Homework: Unsupervised Learning, Dimensionality Reduction, and Anomaly Detection

Objective

The goal of this assignment is to explore the structure of real-world data using unsupervised learning techniques.

Unlike supervised learning, no labels guide the analysis. Your task is to investigate the geometry, hidden structure, clusters, and abnormal behaviors within the dataset.

This assignment combines:

- Clustering methods
- Dimensionality reduction
- Principal Component Analysis (PCA)
- Multi-dimensional anomaly detection
- Critical interpretation of results

You are expected not only to apply algorithms, but also to explain:

- What the algorithms discovered
- Why they behave differently
- What assumptions they make
- What limitations they have

1 Dataset Selection (10%)

Requirements

Choose a dataset satisfying all of the following:

- At least 1000 rows
- At least 8 features
- Mostly numerical variables

- Real-world dataset
- No pre-labeled anomaly column

You may use datasets from:

- Kaggle
- UCI Repository
- OpenML
- Government/open-data repositories
- Cybersecurity logs
- Sensor data
- Financial data
- Medical measurements
- IoT systems

You Must Describe

- Where the dataset comes from
- Why it was collected
- Who collected it (if known)
- What each feature represents
- Possible limitations or biases in the data

2 Exploratory Data Analysis (15%)

Perform a complete exploratory analysis.

Structural Analysis

- Number of rows and columns
- Missing values
- Data types
- Statistical summary
- Correlation matrix

Distribution Analysis

For each major feature, include:

- Histogram
- Boxplot
- Skewness
- Kurtosis

Outlier Exploration

Discuss:

- Heavy tails
- Extreme values
- Possible noise
- Measurement artifacts
- Data-entry problems

3 Dimensionality Reduction (20%)

3.1 Principal Component Analysis (PCA)

Apply PCA to the dataset.

You must include:

- Explained variance ratio
- Cumulative explained variance
- Scree plot
- Projection into:
  - 2D
  - 3D

Discuss:

- How many components are needed
- Information loss
- Correlated variables
- Interpretation of major components

Discuss:

- Local vs global structure
- Cluster separation
- Stability
- Computational cost
- Interpretability

4 Clustering Analysis (20%)

Apply at least three clustering methods.

Possible methods:

- K-Means
- DBSCAN
- Hierarchical Clustering

Clustering on the Feature Space

Apply clustering not only to the samples, but also to the feature space itself (that is, to the transpose of the data matrix).

Discuss the following questions:

- What information can be obtained by clustering the features?
- Did the analysis reveal groups of related or redundant features?
- Were there features that appeared isolated or significantly different from the others?
- Did this analysis provide insights that were not visible when clustering the samples directly?
- In what types of problems or domains might clustering features be particularly useful?

Explain whether clustering the feature space helped uncover hidden structure, dependencies, correlations, or potential feature hierarchies within the dataset.

Required Analysis

For each method:

Explain

- Mathematical intuition
- Assumptions
- Strengths and weaknesses

Evaluate

Use clustering metrics, the average variance in each cluster vs the global variance. Is this metric suitable? If so, explain. Otherwise, provide an alternative.

Visualize

Show clusters on PCA projection.

Discussion

Discuss:

- Why different algorithms produce different clusters
- Sensitivity to hyperparameters
- Cluster geometry
- Density-based vs centroid-based approaches
- Situations where clustering may fail

5 Multi-Dimensional Anomaly Detection (25%)

This is the main part of the assignment.

You must apply and compare the following anomaly detection methods.

5.1 Z-Score Method

Apply a classical Z-score method.

Discuss:

- Assumption of Gaussianity
- Why this approach may fail in high dimensions
- Feature-wise vs global anomaly detection

5.2 Isolation Forest

Apply Isolation Forest.

Analyze:

- Isolation mechanism
- Random partitioning
- Contamination parameter
- Advantages in high-dimensional settings

Visualize detected anomalies.

5.3 Local Outlier Factor (LOF)

Apply LOF.

Discuss:

- Local density estimation
- Neighborhood effects
- Sensitivity to parameter k
- Difference between global and local anomalies

Visualize anomalies.

6 Comparison Section

Compare all anomaly detection methods.

The discussion should address:

- Which anomalies are shared between methods
- Which anomalies are method-specific
- Sensitivity to scaling
- Sensitivity to dimensionality
- Runtime considerations
- Robustness to noise
- Interpretability
- False positives vs false negatives

7 Critical Analysis and Reflection (10%)

Write a discussion section addressing the following topics:

- Why anomaly detection is fundamentally difficult
- Curse of dimensionality
- Why distance becomes problematic in high dimensions
- Difference between noise and anomaly
- The danger of assuming Gaussian distributions
- Why visualization may be misleading after dimensionality reduction

Ethical Considerations

Discuss ethical considerations such as:

- False alarms
- Surveillance
- Medical applications
- Cybersecurity applications

8 Final Visualization Dashboard (Optional Bonus +5%)

Build a small dashboard or notebook section containing:

- PCA visualization
- Cluster visualization
- Interactive anomaly exploration
- Pairplots
- Heatmaps

9 Submission Requirements

Submit the following:

Report

The report must include:

- Methodology
- Mathematical explanations
- Visualizations
- Interpretation
- Critical discussion

This section should be presented inside text boxes within the notebook

Code

Provide:

- Clean notebook or scripts
- Reproducible pipeline
- Comments and explanations
- Meaningful functions and variables names

Important Notes

You are **NOT** graded on obtaining "beautiful" clusters.

You are graded on:

- Depth of understanding
- Critical thinking and analytical reasoning
- Interpretation of results
- Methodological rigor
- Clarity of communication

In real-world data:

- Clusters may not exist
- Anomalies may be ambiguous
- Different methods may disagree

Understanding *why* this happens is an essential part of the assignment.

It is important to note that the inability to identify meaningful clusters, detect significant anomalies, or achieve effective dimensionality reduction does not necessarily indicate poor analysis. In some cases, the underlying structure may simply not exist within the dataset. If this is the case, you should clearly explain your findings and provide a well-reasoned justification based on the characteristics of the data and the methods applied.

Submission Deadline

10.6.2026

Good Luck

Data science is not only about building models.

It is about learning how to reason under uncertainty, incomplete information, and imperfect assumptions.